PANDAS DATA ANALYTICS PROJECT

Vietnam Enterprise Performance 2018-2019

Data Cleaning - Aggregation - Industry Analysis - Segmentation - Time Comparison

Project theme

From raw enterprise survey data to evidence-based business insights

1. Project Overview

Business scenario. You are working as a junior data analyst with a large enterprise survey dataset. Your task is to transform raw data into a reliable analytical dataset, summarize enterprise performance, compare ownership groups and industries, identify high-performing segments, and examine changes between 2018 and 2019.

Central question: How do revenue, profit and employment differ across enterprise types and VSIC 2007 industries, and how did enterprise performance change from 2018 to 2019?

Important scope. The project is descriptive and exploratory. You must not make causal claims. When you report a relationship, use language such as "is associated with", "is higher in this sample", or "differs across groups".

2. Data Files

File	Purpose
Bieu 1A.00_DN 2018.xlsx	Enterprise-level data for 2018.
Bieu 1A.00&Suy rong B_DN 2019.xlsx	Enterprise-level data for 2019.
VSIC 2007.xls	Industry classification lookup used to identify VSIC 2007 divisions.
Enterprise questionnaire / codebook	Use when needed to understand coded variables and definitions.

Industry classification requirement: All industry summaries in this project must be produced at VSIC 2007 Division level (2-digit). Treat industry codes as strings, preserve leading zeros, derive the first two digits, and validate the match rate with the provided VSIC lookup. Never force unmatched codes to a valid industry.

3. Learning Outcomes and Lesson Alignment

Lesson	Required applications in this project
Lesson 1	DataFrame/Series, selection, .loc/.iloc, Boolean filtering, .map(), .apply(), sorting/ranking, descriptive statistics, correlation.
Lesson 2	Loading Excel data, .info(), dtype correction, missing values, duplicates, concat and merge.
Lesson 3	groupby, agg, filter, transform, apply, MultiIndex, pivot_table and Pandas plotting.
Lesson 4	End-to-end analytical workflow, qcut/ranking-based scores, segmentation and segment-level business analysis.
Lesson 5	String normalization with .str and regex; basic time-oriented analysis using year/date variables and time-based indexing.

4. Core Variables

Use the corresponding variables in each yearly file. Variable names may differ slightly between years, so inspect the raw column names before coding.

Concept	Typical variable	How it is used
Enterprise ID	Masothue	Identifier and merge key. Treat as string.
Enterprise name	TenDN	String cleaning and legal-form text analysis.
Enterprise type	LoaihinhKT	Ownership/type grouping.
Main industry	MaNganhSXKD_Chinh	Create VSIC 2007 2-digit division.
Province	MaTinh_Dieutra	Optional regional analysis.
Revenue	TongDoanhthu_DTDN	Mean, median, growth, group analysis.
After-tax profit	Loinhuan_SauThue	Mean, median, profit margin, segmentation.
Employment	SoLaodong_CuoiNam	Average/total employment, size and growth.
Average employment	SoLaodong_BQ	Preferred denominator for revenue per worker if available.

Required analytical rule: Report both mean and median for revenue and profit. Enterprise data are often highly skewed; comparing mean and median is part of the analysis, not an optional extra.

5. Part A - Load, Inspect and Harmonize the Data

A1. Load and inspect each file

- Load the 2018 enterprise file, the 2019 enterprise file and the VSIC 2007 lookup with Pandas.
- Report .shape, column names, .head() and .info() for each dataset.
- Identify which columns are identifiers, categorical variables and numerical measurements.
- Use [] selection, .loc and .iloc at least once and briefly explain the difference between label-based and position-based selection.

A2. Harmonize variables across years

- Create a common set of analysis variables for 2018 and 2019.
- Rename columns when necessary so that both yearly DataFrames use consistent names.
- Add a Year variable to each dataset before combining them.
- Keep raw DataFrames unchanged; perform cleaning on copies.

6. Part B - Data Cleaning and Quality Control

B1. Missing values

- Create a missing-value report containing missing count, missing percentage and dtype for every selected variable.
- Identify the variables with the largest missing percentages.
- Decide variable by variable whether to keep, drop or impute missing observations. Explain your decisions. Do not automatically use fillna(0).

B2. Duplicates and identifiers

- Count exact duplicate rows and duplicate tax IDs within each year.
- Investigate duplicate tax IDs before deleting anything. Explain whether duplicates are data errors or multiple records that require aggregation.

- Convert Masothue to a clean string identifier and remove unwanted spaces or trailing ".0" if needed.

B3. Invalid and extreme values

- Check revenue, profit and employment for impossible or suspicious values (for example negative employment or implausible zeros).
- Use quantiles to identify extreme observations. Do not automatically delete the top 1%.
- Create a short table showing the 10 largest revenue values and the 10 largest absolute profit values.
- Explain which observations you keep or exclude from specific calculations and why.

7. Part C - Overall Enterprise Performance

For each year separately, calculate and present a compact summary table containing:

Indicator	Required statistics
Number of enterprises	Count of unique enterprise IDs
Revenue	Total, mean, median, standard deviation
After-tax profit	Total, mean, median, standard deviation
Employment	Total, mean, median
Revenue per worker	Mean and median of firm-level ratio
Profit margin	Mean and median, using an appropriate revenue denominator

- Compare the mean and median of revenue and profit. What does the difference tell you about the distribution?
- Calculate the correlation matrix for revenue, profit and employment. Identify the strongest relationship and explain why correlation does not imply causation.
- Compare the average of firm-level RevenuePerWorker with the aggregate ratio TotalRevenue / TotalEmployment. Explain why the two measures can differ.

8. Part D - Analysis by Enterprise Type / Ownership

Use the coded enterprise-type variable (for example, LoaihinhKT) and, if appropriate, create a simplified ownership grouping such as State/State-controlled, Domestic non-state and Foreign-related. Keep the original code as well.

Task	Requirement
D1	Use value_counts() to report the number and percentage of enterprises in each type.
D2	Use groupby().agg() to calculate firm count, mean/median revenue, mean/median profit, total/mean employment, median revenue per worker and median profit margin by type.
D3	Rank enterprise types by median revenue, median profit and total employment.
D4	Use groupby().transform() to create TypeAverageRevenue and a Boolean variable AboveTypeAverageRevenue for each enterprise.
D5	Use groupby().rank() to identify the top 5 revenue enterprises within each type.
D6	Use groupby().filter() to keep only enterprise types with at least a reasonable number of firms. State and justify your threshold.

9. Part E - VSIC 2007 Industry Analysis (2-digit Division)

E1. Build a valid 2-digit industry variable

- Convert the main industry code to string and normalize it with .str methods.
- Extract the first two digits to create VSIC2.
- Use the provided VSIC 2007 file to attach the 2-digit industry description.
- Report the number and percentage of enterprise records that successfully match the VSIC lookup.
- Keep unmatched codes as "Unmatched" and investigate the most common unmatched values.

E2. Calculate industry-level indicators

For every VSIC2 division, calculate:

- Number of enterprises;
- Mean and median revenue;

- Mean and median after-tax profit;
- Total and average employment;
- Median revenue per worker;
- Median profit margin.

Small-group warning: Do not rank an industry as "best" solely because it has a very high mean based on only a few firms. For ranking exercises, first require a minimum number of enterprises per VSIC2 division (for example 30 or 50), and report the threshold you use.

- Identify the 10 VSIC2 divisions with the highest median revenue.
- Identify the 10 divisions with the highest median after-tax profit.
- Identify the 10 divisions with the largest total employment.
- Identify the industries with the highest median revenue per worker.
- Compare whether the ranking changes when using mean instead of median.

10. Part F - Advanced Group Operations, MultiIndex and Pivot Tables

- Create a MultiIndex summary using Year and OwnershipGroup, and a second one using Year and VSIC2.
- Use unstack() to convert at least one MultiIndex result from long to wide format.
- Create a pivot table with OwnershipGroup as rows, Year as columns and median revenue as values.
- Create a second pivot table with VSIC2 as rows, Year as columns and median profit as values.
- Use a custom groupby().apply() operation to return the top 3 revenue enterprises within each VSIC2 division.
- Create an IndustryAverageRevenue variable with transform() and calculate each firm's revenue gap from its industry average.

11. Part G - Enterprise Performance Segmentation

Purpose. Adapt the quantile-scoring and segmentation logic from the RFM case study to enterprise performance. This is not an RFM model; it is a cross-sectional enterprise segmentation exercise using the same Pandas techniques.

- Create RevenueScore from 1 to 5 using pd.qcut(). Higher revenue receives a higher score.
- Create ProfitScore from 1 to 5 using pd.qcut(). Higher profit receives a higher score. If duplicate boundaries cause problems, use rank(method="first") before qcut.
- Create EmploymentScore from 1 to 5 using pd.qcut(). Higher employment receives a higher score.
- Optionally create ProductivityScore based on RevenuePerWorker.
- Combine the scores into a readable code or segmentation rule.
- Design at least four enterprise segments and justify the rules. Suggested labels include Top Performers, Large Employers, Efficient Smaller Firms, and Vulnerable/Low-Performance Firms.
- Use groupby().agg() to summarize the number of enterprises, mean/median revenue, profit, employment and productivity in each segment.
- Calculate the percentage of enterprises in each segment and propose one practical business/policy interpretation for each segment.

12. Part H - String Manipulation and Regex

- Create CleanName from enterprise name using .str.strip() and a consistent case convention.
- Use .str.contains() to identify names containing selected legal-form keywords (for example CO PHAN, TNHH, DNTN or HTX, depending on the text format in the file).
- Use .str.extract() with a regular expression to create a TextLegalForm variable from enterprise names.
- Compare TextLegalForm with the coded enterprise-type variable. Report mismatches as a data-quality diagnostic; do not assume the text rule is always correct.
- Use string operations to standardize the VSIC code before extracting the 2-digit division.

13. Part I - 2018-2019 Time-Based Comparison

Scope: The dataset contains only two annual observations, so this section is a basic time comparison, not a forecasting exercise.

- After harmonizing 2018 and 2019, combine the two yearly DataFrames with pd.concat().
- Create an annual date variable using pd.to_datetime() and demonstrate a DatetimeIndex on an aggregated yearly summary.
- Calculate yearly total/mean/median revenue, profit and employment and show the change from 2018 to 2019.

- Match enterprises observed in both years by Masothue and calculate firm-level RevenueGrowthPct and EmploymentGrowthPct. For profit, use a level change or another carefully justified measure because percentage growth is problematic when the base profit is zero or negative.
- Calculate the median revenue growth of continuing firms by OwnershipGroup.
- Calculate the median revenue growth of continuing firms by VSIC2 division.
- Identify the 10 divisions with the strongest median revenue growth, subject to a minimum number of continuing firms.
- Explain survivorship bias: firms observed in both years are not necessarily representative of all enterprises.

14. Part J - Visualization

Create at least five clear charts using Pandas/Matplotlib. Every chart must have a title, axis labels, readable categories and a short interpretation.

Chart	Required topic
1	Enterprise count by ownership/type
2	Median revenue and/or median profit by ownership group
3	Top 10 VSIC2 divisions by median revenue
4	Top 10 VSIC2 divisions by total employment
5	2018 vs 2019 summary for revenue, profit or employment
Optional	Scatter plot of revenue vs employment; for visualization only, you may filter extreme values using a clearly stated quantile rule.

15. Final Business Questions

Your notebook must end with concise, evidence-based answers to the following questions. Support each answer with a table, statistic or chart.

1. How did average and median revenue, profit and employment change from 2018 to 2019?
2. Which enterprise type/ownership group has the highest median revenue and highest median profit? Is it also the largest employer?
3. Which VSIC 2007 2-digit industries appear strongest when performance is measured by median revenue, median profit and revenue per worker?
4. Which industries employ the largest number of workers, and are these also the industries with the highest revenue per worker?
5. How different are conclusions based on the mean compared with the median? What role do outliers play?
6. Among enterprises observed in both years, which ownership groups and VSIC2 divisions show the strongest median revenue growth?
7. What are the main characteristics of the enterprise performance segments created in Part G?
8. What is the relationship among revenue, profit and employment? State clearly why the correlation cannot be interpreted as causation.

16. Deliverables

Deliverable	Requirement
1. Jupyter Notebook	A fully reproducible .ipynb file with code, outputs, tables, charts and written interpretations.
2. Clean analytical dataset	A CSV or Parquet file containing the harmonized variables used in the final analysis.
3. Executive Summary	500-700 words at the end of the notebook: data issues, five or more findings, limitations and recommendations.
4. Reproducibility	The notebook must run from top to bottom without manual editing of intermediate results.

17. Minimum Pandas Requirements

Your notebook must demonstrate the following methods/ideas at least once in a meaningful context:

[ ] head / shape / info	[ ] [] selection, .loc and .iloc
[ ] Boolean masks and .isin()	[ ] isnull / dropna or justified fillna
[ ] drop_duplicates	[ ] astype / pd.to_numeric

[ ] string methods with .str	[ ] at least one regex method
[ ] sort_values and rank	[ ] map and apply
[ ] concat and merge	[ ] groupby with agg
[ ] groupby filter	[ ] groupby transform
[ ] groupby apply	[ ] MultiIndex and unstack
[ ] pivot_table	[ ] describe and corr
[ ] qcut	[ ] pd.to_datetime
[ ] Pandas/Matplotlib plot	

18. Assessment Rubric - 100 Marks

Component	Marks	What is assessed
Data loading and harmonization	10	Correct imports, common variables, identifiers, reproducible structure.
Cleaning and data-quality decisions	15	Missing values, duplicates, dtypes, invalid/extreme values, documented decisions.
Overall performance indicators	10	Correct revenue/profit/employment statistics, ratios and interpretation.
Enterprise type analysis	10	GroupBy/Agg, transform, rank/filter and meaningful comparison.
VSIC 2007 2-digit analysis	15	String code cleaning, lookup merge/validation, industry statistics and ranking.
Advanced group operations and pivots	10	MultiIndex, unstack, pivot_table, group apply/transform.
Segmentation	10	qcut/rank scores, defensible segment rules and segment analysis.
String and time-based analysis	10	Text normalization/regex plus 2018-2019 comparison and firm matching.
Visualization and business interpretation	5	Readable charts and evidence-based insights.
Executive summary and reproducibility	5	Clear final story, limitations, notebook runs from top to bottom.
TOTAL	100	

19. Analytical Quality Rules

- Do not delete observations only because they look large. Investigate first.
- Do not replace all missing values with zero.
- Do not calculate means of identifiers such as tax IDs or industry codes.
- Do not convert industry codes to integers if doing so destroys leading zeros.
- Do not rank tiny industry groups against large groups without a minimum-count rule.
- Do not calculate percentage profit growth without considering zero or negative base profit.
- Do not call a correlation a causal effect.
- Every major table or chart must be followed by a short interpretation in plain English.

Submission naming convention: StudentID_FullName_Enterprise_Project.ipynb and StudentID_FullName_Enterprise_Clean.csv

Appendix A - Suggested Derived Variables

Variable	Suggested definition / note
VSIC2	First two digits of the standardized main industry code.
RevenuePerWorker	Revenue / average employment; if average employment is unavailable, use year-end employment and state this.
ProfitMargin	After-tax profit / an appropriate revenue measure; set undefined ratios to missing rather than forcing zero.
AboveTypeAverageRevenue	Revenue > mean revenue within enterprise type.
AboveIndustryAverageRevenue	Revenue > mean revenue within VSIC2 division.
RevenueGrowthPct	100 x (Revenue2019 - Revenue2018) / Revenue2018, only when the 2018 denominator is valid and positive.

EmploymentGrowthPct	100 x (Employment2019 - Employment2018) / Employment2018, only when the 2018 denominator is positive.
RevenueScore / ProfitScore / EmploymentScore	1-5 quantile scores using qcut; use rank first when duplicate boundaries prevent qcut.
EnterpriseSegment	A student-defined segment based on the quantile scores, with rules explained in the notebook.

END OF PROJECT BRIEF